Lesson 6.1 — Exponential Functions

Big idea: $f(x) = a \cdot b^x$ grows or decays by a constant factor every step. The base b decides whether you’re growing ($b > 1$) or decaying ($0 < b < 1$).

Quick review

Form: $f(x) = a \cdot b^x$ with $a \neq 0$, $b > 0$, $b \neq 1$.
Initial value: $a = f(0)$.
Base: $b > 1 \to$ growth; $0 < b < 1 \to$ decay.
Domain / range: domain ■; range $(0, \infty)$ if $a > 0$, $(-\infty, 0)$ if $a < 0$.
Horizontal asymptote: $y = 0$ (the function approaches but never crosses).

Worked example

Worked example. Sketch $f(x) = 3 \cdot 2^x$. State a, b, intercept, end behavior.

$a = 3$, $b = 2$ (growth). y-intercept $(0, 3)$. As $x \to -\infty$, $f \to 0$. As $x \to +\infty$, $f \to \infty$. Doubles every unit increase in x.

Warm-ups

1. Identify a and b for $f(x) = 5 \cdot 3^x$.
2. Is $f(x) = 4 \cdot (1/2)^x$ growth or decay?
3. Find $f(0)$ for $f(x) = 7 \cdot 2^x$.

Practice

1. Sketch (describe) $f(x) = 2 \cdot 4^x$. State y-intercept and HA.
2. Sketch (describe) $f(x) = 6 \cdot (0.5)^x$. State y-intercept and HA.
3. Find $f(3)$ for $f(x) = 5 \cdot 2^x$.

Challenge

1. An exponential function passes through $(0, 4)$ and $(2, 36)$. Find a and b.
2. Why does the graph of any $f(x) = a \cdot b^x$ never cross its horizontal asymptote $y = 0$? (For $a > 0$.)

Answer key — Lesson 6.1

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $a = 5$, $b = 3$.
2. Decay ($b = 1/2 < 1$).
3. 7.

Practice

1. y-intercept (0, 2). HA $y = 0$. Quadruples every step.
2. y-intercept (0, 6). HA $y = 0$. Halves every step (decreasing).
3. $5 \cdot 8 = 40$.

Challenge

1. $f(0) = a = 4$. $f(2) = 4b^2 = 36 \Rightarrow b^2 = 9 \Rightarrow b = 3$. $f(x) = 4 \cdot 3^x$.
2. Because $b^x > 0$ for any real x and any $b > 0$, multiplying by positive a keeps $f(x) > 0$. The product can get arbitrarily close to 0 but never equals it.